

HDOC Day-17

Question: Menu-Driven Program for Digit Operations Using User-Defined Functions

1. Problem Statement & Identification

- **Problem:** Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer using separate user-defined functions (void return type and no parameters) for each option:
 1. Generate the largest possible number using its digits.
 2. Generate the smallest possible number using its digits.
 3. Swap its first and last digits.
 4. Check whether it is a palindrome number.
- **Input:** Menu choice (choice), and inside each function, a positive integer (num).
- **Output:** Display result directly inside the corresponding user-defined function.

2. Algorithm

1. **Start**
2. Display menu options (1 to 4).

3. Read user choice.
4. Execute user-defined function based on choice using switch-case:

- **Case 1: generateLargestNumber()**

1. Read num. Count digit frequencies in freq[10].
2. Build largest by appending digits from index 9 down to 0.
3. Display largest.

- **Case 2: generateSmallestNumber()**

1. Read num. Count digit frequencies in freq[10].
2. Set first non-zero digit as the leading digit of smallest.
3. Append remaining digits in ascending order from index 0 to 9.
4. Display smallest.

- **Case 3: swapFirstAndLastDigits()**

1. Read num. Extract last_digit = num % 10.
2. Find first_digit and number of middle digits.
3. Construct swapped_num = (last_digit * 10^{count}) + middle + first_digit.

4. Display swapped_num.

- **Case 4: checkPalindrome()**

1. Read num. Reverse num into reversed.

2. If num == reversed, print "It is a palindrome number", else print "It is not a palindrome number".

- **Default:** Print "Invalid Choice!".

5. End

3. Test Case Table

Test Case	Choice	Input Number	Operation	Expected Output	Actual Output	Status
1	1	38195	Generate Largest Number	98531	98531	Pass
2	2	50421	Generate Smallest Number	10245	10245	Pass

Test Case	Choice	Input Number	Operation	Expected Output	Actual Output	Status
			st Number			
3	3	12345	Swap First and Last Digits	52341	52341	Pass
4	4	12321	Check Palindrome Number	It is a palindrome number	It is a palindrome number	Pass

4. C Source Code

```
#include <stdio.h>
#include <math.h>

// Function Prototypes (void return type, no parameters)
void generateLargestNumber(void);
void generateSmallestNumber(void);
void swapFirstAndLastDigits(void);
void checkPalindrome(void);
```

```
int main(void) {
    int choice;

    printf("--- MENU ---\n");
    printf("1. Generate the largest possible number using its digits\n");
    printf("2. Generate the smallest possible number using its digits\n");
    printf("3. Swap its first and last digits\n");
    printf("4. Check whether it is a palindrome number\n");
    printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            generateLargestNumber();
            break;
        case 2:
            generateSmallestNumber();
            break;
        case 3:
            swapFirstAndLastDigits();
            break;
        case 4:
            checkPalindrome();
            break;
        default:
            printf("\nInvalid Choice!\n");
            break;
    }

    return 0;
}

void generateLargestNumber(void) {
    int num, temp, digit;
    int freq[10] = {0};
    long long largest = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);
```

```
temp = num;
while (temp > 0) {
    digit = temp % 10;
    freq[digit]++;
    temp /= 10;
}

for (int i = 9; i >= 0; i--) {
    while (freq[i] > 0) {
        largest = largest * 10 + i;
        freq[i]--;
    }
}

printf("\nNumber: %d, Largest Possible Number: %lld\n", num, largest);
}

void generateSmallestNumber(void) {
    int num, temp, digit;
    int freq[10] = {0};
    long long smallest = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        freq[digit]++;
        temp /= 10;
    }

    for (int i = 1; i <= 9; i++) {
        if (freq[i] > 0) {
            smallest = i;
            freq[i]--;
            break;
        }
    }
}
```

```

for (int i = 0; i <= 9; i++) {
    while (freq[i] > 0) {
        smallest = smallest * 10 + i;
        freq[i]--;
    }
}

printf("\nNumber: %d, Smallest Possible Number: %lld\n", num,
smallest);
}

void swapFirstAndLastDigits(void) {
    int num, temp, first_digit, last_digit, digits_count = 0;
    long long swapped_num;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num < 10) {
        printf("\nNumber: %d, Swapped Number: %d\n", num, num);
        return;
    }

    last_digit = num % 10;
    temp = num;

    while (temp >= 10) {
        temp /= 10;
        digits_count++;
    }
    first_digit = temp;

    int pow10 = 1;
    for (int i = 0; i < digits_count; i++) {
        pow10 *= 10;
    }

    int middle = (num % pow10) / 10;
    swapped_num = (long long)last_digit * pow10 + middle * 10 + first_digit;

```

```
    printf("\nNumber: %d, Swapped Number: %lld\n", num,
swapped_num);
}

void checkPalindrome(void) {
    int num, temp, digit;
    long long reversed = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        reversed = reversed * 10 + digit;
        temp /= 10;
    }

    if (num == reversed) {
        printf("\nNumber: %d, Output: It is a palindrome number\n", num);
    } else {
        printf("\nNumber: %d, Output: It is not a palindrome number\n",
num);
    }
}
```

6. Compilation, Execution & Screenshots



```
kali@kali - /HDQC_Day_17
Session Actions Edit View Help
kali@kali ~$ ls
main.c HDQC_Day_17
cd HDQC_Day_17
kali@kali ~/HDQC_Day_17$ ls
main.c q1.c
kali@kali ~/HDQC_Day_17$ gcc q1.c -o q1
./q1
MENU
1. Generate the largest possible number using its digits
2. Generate the smallest possible number using its digits
3. Swap its first and last digits
4. Check whether it is a palindrome number
Enter your choice (1-4): 1
Enter a positive integer: 38195
Number: 38195, Largest Possible Number: 98531
kali@kali ~/HDQC_Day_17$
```